Grade 11 – Term 1
C8: Student's t in Context

Contents

1 Models and calculation tools 1

2 Direct probability problems 2

3 Inverse probability problems 10

Evaluated criterion

C8 Calculate contextual t probabilities and find cutoffs from target probabilities.

1 Models and calculation tools

Open the normal and Student's t reference tables.

Page 1 gives $\Phi(z) = P(Z \leq z)$ for positive z on a 0.01 grid (through 2.59). Page 2 gives positive t critical values for upper-tail probabilities $0.30, 0.20, 0.10, 0.05, 0.025, 0.01, 0.005$, for df 1–30 and selected larger df. Those entries do not give arbitrary CDF values. Use a suitable calculator's t CDF and inverse t CDF for this activity; normal CDF and inverse normal are also allowed. Numerical CDF and quantile results in the solutions are supplementary to the tables. Never turn table bounds into a single probability or error.

Write $F_\nu(t) = P(T \leq t)$ for $T \sim t_\nu$. Symmetry gives $F_\nu(-c) = 1 - F_\nu(c)$; a right tail is $1 - F_\nu(c)$. An interval has probability $F_\nu(b) - F_\nu(a)$ and its outside complement has probability $F_\nu(a) + 1 - F_\nu(b)$. Replace F_ν by Φ for normal calculations. Endpoints have 0 probability. For central area p, each tail has $(1-p)/2$; for combined symmetric tails q, each tail has $q/2$.

A contextual location-scale model is explicitly $X = m + aT$, with $a > 0$ a scale parameter; a is not automatically the standard deviation. Thus $T = (X - m)/a$. These are stipulated models for signed deviations, not claims that arbitrary individual measurements follow t. For independent random observations from a normal population, with unknown population standard deviation, the sampling statistic instead is

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Here $\bar{X}$ and S vary across samples; do not treat observed s as a fixed population standard deviation.

Attempt each task before checking the worked solution. Calculate with full precision, then report probabilities to 4 decimals, standardized quantiles to 3 decimals, contextual values to 2 decimals, and percentage errors to 2 decimals.

Displayed rounded intermediate values are not the inputs used for errors. Diagrams shade the requested region; curves continue beyond the displayed range.

2 Direct probability problems

Problems:

2.1. Delivery schedule deviations
2.2. Production cost deviations
2.3. Electricity forecast deviations
2.4. Monthly sales deviations
2.5. Commute forecast deviations
2.6. Revenue planning alerts

2.1 Delivery schedule deviations

Problem. A planning team describes how far delivery time falls below or rises above its scheduled time, in minutes. Use the chance pattern identified by the row marked 9 on page 2 of the supplied reference tables. Its middle represents exactly meeting the scheduled time. Each step of 1 in that pattern represents 5 minutes here. Amounts below the middle mean less than the scheduled time, and amounts above it mean more.

Questions/Tasks

C8 Find the chance of arriving before 7.5 minutes after the scheduled time, and the chance of arriving more than 6 minutes early. Show how to express each requested outcome using the reference pattern. Explain how matching chances on either side of the middle or the chance of all other outcomes helps, where needed, and interpret each answer.

Worked solution

Translating the information into mathematics Let X be delivery time minus its scheduled time, in minutes, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 9 degrees of freedom. The middle and conversion give

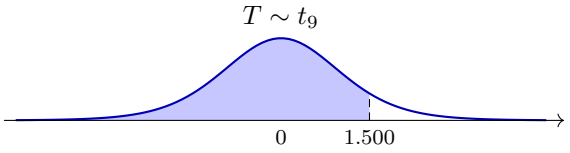
$$X = 0 + 5T, \qquad T = \frac{X - 0}{5}, \qquad T \sim t_9.$$

The location is 0 and the scale is 5 minutes. This scale is not automatically a population standard deviation.

Standardize each boundary using $(x - 0)/5$.

Part 1.

$$P(X < 7.5) = P(T < 1.5).$$



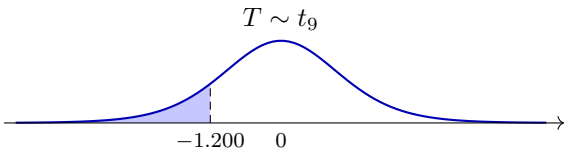
For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. The region is cumulative area to the left. Supplementary CDF values: $F_9(1.5) \approx 0.9160747$.

$$P(T < 1.5) = F_9(1.5) \approx 0.9161.$$

Interpretation. About 91.61% of deliveries arrive before 7.5 minutes after their scheduled time.

Part 2.

$$P(X < -6) = P(T < -1.2).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. The region is cumulative area to the left. By symmetry, $F_9(-1.2) = 1 - F_9(1.2) \approx 1 - 0.8696134 = 0.1303866$. Supplementary CDF values: $F_9(-1.2) \approx 0.1303866$.

$$P(T < -1.2) = F_9(-1.2) \approx 0.1304.$$

Interpretation. About 13.04% of deliveries arrive more than 6 minutes early.

2.2 Production cost deviations

Problem. A planning team describes how far batch cost falls below or rises above its budget, in monetary units. Use the chance pattern identified by the row marked 14 on page 2 of the supplied reference tables. Its middle represents exactly meeting the budget. Each step of 1 in that pattern represents 20 monetary units here. Amounts below the middle mean less than the budget, and amounts above it mean more.

Questions/Tasks

C8 Find the chance of costing more than 30 monetary units above budget, and the chance of costing more than the budget reduced by 16 monetary units. Show how to express each requested outcome using the reference pattern. Explain how matching chances on either side of the middle or the chance of all other outcomes helps, where needed, and interpret each answer.

Worked solution

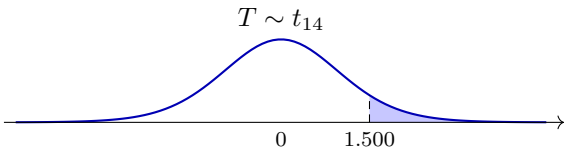
Translating the information into mathematics Let X be batch cost minus its budget, in monetary units, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 14 degrees of freedom. The middle and conversion give

$$X = 0 + 20T, \qquad T = \frac{X - 0}{20}, \qquad T \sim t_{14}.$$

The location is 0 and the scale is 20 monetary units. This scale is not automatically a population standard deviation. Standardize each boundary using $(x - 0)/20$.

Part 1.

$$P(X > 30) = P(T > 1.5).$$



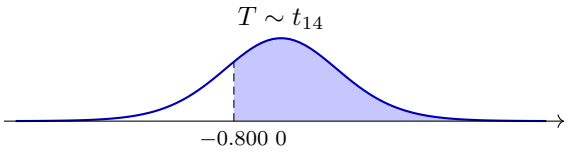
For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Take the complement of the cumulative area. Supplementary CDF values: $F_{14}(1.5) \approx 0.9220873$.

$$P(T > 1.5) = 1 - F_{14}(1.5) \approx 0.0779.$$

Interpretation. About 7.79% of batches cost more than 30 monetary units above budget.

Part 2.

$$P(X > -16) = P(T > -0.8).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Take the complement of the cumulative area. By symmetry, $F_{14}(-0.8) = 1 - F_{14}(0.8) \approx 1 - 0.7814622 = 0.2185378$. Supplementary CDF values: $F_{14}(-0.8) \approx 0.2185378$.

$$P(T > -0.8) = 1 - F_{14}(-0.8) \approx 0.7815.$$

Interpretation. About 78.15% of batches cost more than the budget reduced by 16 monetary units.

2.3 Electricity forecast deviations

Problem. A planning team describes how far household electricity use falls below or rises above its forecast, in kilowatt-hours. Use the chance pattern identified by the row marked 19 on page 2 of the supplied reference tables. Its middle represents exactly meeting the forecast. Each step of 1 in that pattern represents 50 kilowatt-hours here. Amounts below the middle mean less than the forecast, and amounts above it mean more.

Questions/Tasks

C8 Find the chance of using more than 12.5 but less than 57.5 kilowatt-hours above forecast. Show how to express each requested outcome using the reference pattern. Explain how matching chances on either side of the middle or the chance of all other outcomes helps, where needed, and interpret each answer.

Worked solution

Translating the information into mathematics Let X be household electricity use minus its forecast, in kilowatt-hours, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 19 degrees of freedom. The middle and conversion give

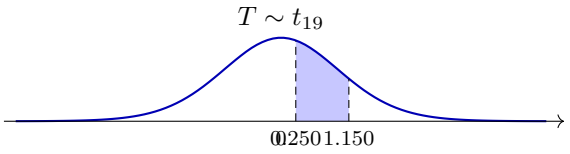
$$X = 0 + 50T, \qquad T = \frac{X - 0}{50}, \qquad T \sim t_{19}.$$

The location is 0 and the scale is 50 kilowatt-hours. This scale is not automatically a population standard deviation.

Standardize each boundary using $(x - 0)/50$.

Part 1.

$$P(12.5 < X < 57.5) = P(0.25 < T < 1.15).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Subtract the left cumulative area from the right cumulative area. Supplementary CDF values: $F_{19}(0.25) \approx 0.5973648, \quad F_{19}(1.15) \approx 0.8677933.$

$$P(0.25 < T < 1.15) = F_{19}(1.15) - F_{19}(0.25) \approx 0.2704.$$

Interpretation. About 27.04% of households use more than 12.5 but less than 57.5 kWh above forecast

2.4 Monthly sales deviations

Problem. A planning team describes how far monthly sales falls below or rises above its target, in units. Use the chance pattern identified by the row marked 11 on page 2 of the supplied reference tables. Its middle represents exactly meeting the target. Each step of 1 in that pattern represents 20 units here. Amounts below the middle mean less than the target, and amounts above it mean more.

Questions/Tasks

C8 Find the chance of falling short of target by more than 17 but less than 36 units. Show how to express each requested outcome using the reference pattern. Explain how matching chances on either side of the middle or the chance of all other outcomes helps, where needed, and interpret each answer.

Worked solution

Translating the information into mathematics Let X be monthly sales minus its target, in units, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 11 degrees of freedom. The middle and conversion give

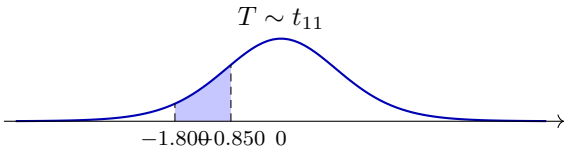
$$X = 0 + 20T, \qquad T = \frac{X - 0}{20}, \qquad T \sim t_{11}.$$

The location is 0 and the scale is 20 units. This scale is not automatically a population standard deviation.

Standardize each boundary using $(x - 0)/20$.

Part 1.

$$P(-36 < X < -17) = P(-1.8 < T < -0.85).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Subtract the left cumulative area from the right cumulative area. By symmetry, $F_{11}(-1.8) = 1 - F_{11}(1.8) \approx 1 - 0.9503421 = 0.0496579$. By symmetry, $F_{11}(-0.85) = 1 - F_{11}(0.85) \approx 1 - 0.7932783 = 0.2067217$. Supplementary CDF values: $F_{11}(-1.8) \approx 0.0496579, \quad F_{11}(-0.85) \approx 0.2067217$.

$$P(-1.8 < T < -0.85) = F_{11}(-0.85) - F_{11}(-1.8) \approx 0.1571.$$

Interpretation. About 15.71% of monthly sales outcomes fall short of target by more than 17 but less than 36 units.

2.5 Commute forecast deviations

Problem. A planning team describes how far commute time falls below or rises above its forecast, in minutes. Use the chance pattern identified by the row marked 24 on page 2 of the supplied reference tables. Its middle represents exactly meeting the forecast. Each step of 1 in that pattern represents 10 minutes here. Amounts below the middle mean less than the forecast, and amounts above it mean more.

Questions/Tasks

C8 Find the chance of taking more than the forecast reduced by 9.5 minutes, but less than the forecast increased by 10.5 minutes. Show how to express each requested outcome using the reference pattern. Explain how matching chances on either side of the middle or the chance of all other outcomes helps, where needed, and interpret each answer.

Worked solution

Translating the information into mathematics Let X be commute time minus its forecast, in minutes, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 24 degrees of freedom. The middle and conversion give

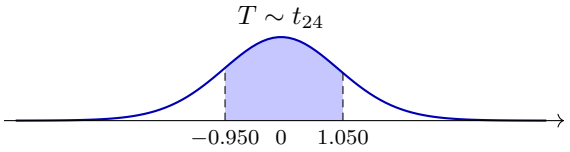
$$X = 0 + 10T, \qquad T = \frac{X - 0}{10}, \qquad T \sim t_{24}.$$

The location is 0 and the scale is 10 minutes. This scale is not automatically a population standard deviation.

Standardize each boundary using $(x - 0)/10$.

Part 1.

$$P(-9.5 < X < 10.5) = P(-0.95 < T < 1.05).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Subtract the left cumulative area from the right cumulative area. By symmetry, $F_{24}(-0.95) = 1 - F_{24}(0.95) \approx 1 - 0.8242092 = 0.1757908$. Supplementary CDF values: $F_{24}(-0.95) \approx 0.1757908$, $F_{24}(1.05) \approx 0.8479116$.

$$P(-0.95 < T < 1.05) = F_{24}(1.05) - F_{24}(-0.95) \approx 0.6721.$$

Interpretation. About 67.21% of commutes take more than the forecast reduced by 9.5 minutes and less than the forecast increased by 10.5 minutes.

2.6 Revenue planning alerts

Problem. A planning team describes how far weekly revenue falls below or rises above its plan, in units of 1,000 monetary units. Use the chance pattern identified by the row marked 5 on page 2 of the supplied reference tables. Its middle represents exactly meeting the plan. Each step of 1 in that pattern represents 8,000 monetary units here. Amounts below the middle mean less than the plan, and amounts above it mean more.

Questions/Tasks

C8 Find the chance of revenue being more than 8,400 monetary units below plan or more than 10,800 monetary units above plan. Also find the chance of revenue being more than 16,000 monetary units below or above plan. Show how to express each requested outcome using the reference pattern. Explain how matching chances on either side of the middle or the chance of all other outcomes helps, where needed, and interpret each answer.

Worked solution

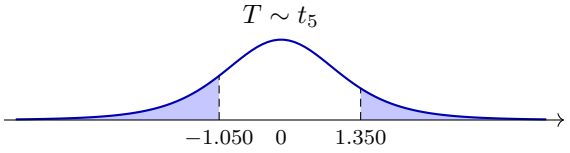
Translating the information into mathematics Let X be weekly revenue minus its plan, in units of 1,000 monetary units, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 5 degrees of freedom. The middle and conversion give

$$X = 0 + 8T, \qquad T = \frac{X - 0}{8}, \qquad T \sim t_5.$$

The location is 0 and the scale is 8,000 monetary units. This scale is not automatically a population standard deviation. Standardize each boundary using $(x - 0)/8$.

Part 1.

$$P(X < -8.4 \text{ or } X > 10.8) = P(T < -1.05 \text{ or } T > 1.35).$$



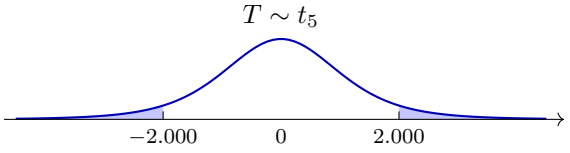
For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Add the disjoint outside regions, or subtract the intervening interval from 1. By symmetry, $F_5(-1.05) = 1 - F_5(1.05) \approx 1 - 0.8291023 = 0.1708977$. Supplementary CDF values: $F_5(-1.05) \approx 0.1708977$, $F_5(1.35) \approx 0.8825456$.

$$P(T < -1.05 \text{ or } T > 1.35) = F_5(-1.05) + 1 - F_5(1.35) \approx 0.2884.$$

Interpretation. About 28.84% of weekly revenue outcomes are more than 8,400 monetary units below plan or more than 10,800 monetary units above plan.

Part 2.

$$P(X < -16 \text{ or } X > 16) = P(T < -2 \text{ or } T > 2).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Add the disjoint outside regions, or subtract the intervening interval from 1. By symmetry, $F_5(-2) = 1 - F_5(2) \approx 1 - 0.9490303 = 0.0509697$. Supplementary CDF values:

$F_5(-2) \approx 0.0509697, \quad F_5(2) \approx 0.9490303.$

$$P(T < -2 \text{ or } T > 2) = F_5(-2) + 1 - F_5(2) \approx 0.1019.$$

Interpretation. About 10.19% of weekly revenue outcomes are more than 16,000 monetary units below or above plan.



3 Inverse probability problems

Problems:

3.1. Top delivery delays

3.2. Bottom cost deviations

3.3. Central energy deviations

3.4. Central sales deviations

3.5. 2-tail checks on filling weights

3.6. 2-tail checks on processing times

3.1 Top delivery delays

Problem. A planning team describes how far delivery time falls below or rises above its scheduled time, in minutes. Use the chance pattern identified by the row marked 9 on page 2 of the supplied reference tables. Its middle represents exactly meeting the scheduled time. Each step of 1 in that pattern represents 5 minutes here. Amounts below the middle mean less than the scheduled time, and amounts above it mean more.

Questions/Tasks

C8 Find how many minutes late a delivery must be so that only 5 percent of deliveries arrive later. Express the requested outcome using the reference pattern, work out the share at or below each boundary, find the boundaries in that pattern, and convert to the original units. Explain how the shares above or below the boundaries are used and interpret the result.

Worked solution

Translating the information into mathematics Let X_T be delivery time minus its scheduled time, in minutes, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 9 degrees of freedom. The middle and conversion give

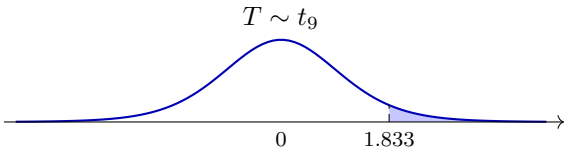
$$X_T = 0 + 5T, \qquad T = \frac{X_T - 0}{5}, \qquad T \sim t_9.$$

The location is 0 and the scale is 5 minutes. This scale is not automatically a population standard deviation.

$$P(X_T > x) = 0.0500 \quad \Longrightarrow \quad P(T \le c) = 0.9500, \qquad c = \frac{x - 0}{5}.$$

A top percentage is a right tail, so subtract it from 1 before inverse CDF.

$$c_T = F_9^{-1}(0.95) \approx 1.833.$$



$$x_T = 0 + 5c_T \approx 9.17 \text{ minutes.}$$

Interpretation. About 5% of deliveries are above 9.17 minutes.

3.2 Bottom cost deviations

Problem. A planning team describes how far batch cost falls below or rises above its budget, in monetary units. Use the chance pattern identified by the row marked 11 on page 2 of the supplied reference tables. Its middle represents exactly meeting the budget. Each step of 1 in that pattern represents 20 monetary units here. Amounts below the middle mean less than the budget, and amounts above it mean more.

Questions/Tasks

C8 Find the amount compared with budget below which 10 percent of batch costs fall. Express the requested outcome using the reference pattern, work out the share at or below each boundary, find the boundaries in that pattern, and convert to the original units. Explain how the shares above or below the boundaries are used and interpret the result.

Worked solution

Translating the information into mathematics Let X_T be batch cost minus its budget, in monetary units, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 11 degrees of freedom. The middle and conversion give

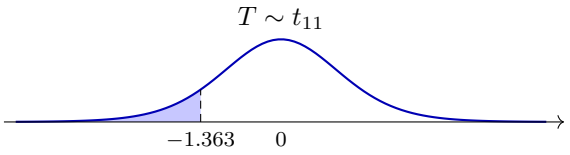
$$X_T = 0 + 20T, \qquad T = \frac{X_T - 0}{20}, \qquad T \sim t_{11}.$$

The location is 0 and the scale is 20 monetary units. This scale is not automatically a population standard deviation.

$$P(X_T < x) = 0.1000 \quad \Longrightarrow \quad P(T \le c) = 0.1000, \qquad c = \frac{x - 0}{20}.$$

The bottom percentage is already a left cumulative probability; the cutoff is negative by symmetry.

$$c_T = F_{11}^{-1}(0.1) \approx -1.363.$$



$$x_T = 0 + 20c_T \approx -27.27 \text{ monetary units.}$$

Interpretation. About 10% of batches cost more than 27.27 monetary units below budget.

3.3 Central energy deviations

Problem. A planning team describes how far electricity use falls below or rises above its forecast, in kilowatt-hours. Use the chance pattern identified by the row marked 19 on page 2 of the supplied reference tables. Its middle represents exactly meeting the forecast. Each step of 1 in that pattern represents 50 kilowatt-hours here. Amounts below the middle mean less than the forecast, and amounts above it mean more.

Questions/Tasks



C8 Find 2 amounts equally far below and above forecast enclosing the middle 95 percent of electricity-use outcomes. Express the requested outcome using the reference pattern, work out the share at or below each boundary, find the boundaries in that pattern, and convert to the original units. Explain how the shares above or below the boundaries are used and interpret the result.

Worked solution

Translating the information into mathematics Let X_T be electricity use minus its forecast, in kilowatt-hours, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 19 degrees of freedom. The middle and conversion give

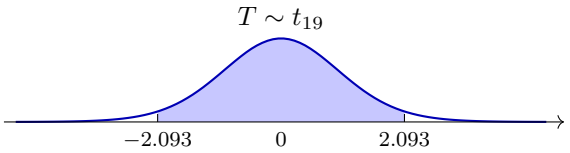
$$X_T = 0 + 50T, \qquad T = \frac{X_T - 0}{50}, \qquad T \sim t_{19}.$$

The location is 0 and the scale is 50 kilowatt-hours. This scale is not automatically a population standard deviation.

$$P(-c < T < c) = 0.9500 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0250.$$

Divide the outside complement $1 - p$ by 2 because symmetry makes the tails equal.

$$F_{19}(c) = 0.9750, \qquad c_T = F_{19}^{-1}(0.975) \approx 2.093.$$



The standardized endpoints are $[-2.093, 2.093]$.

$$L_T = 0 - 50c_T \approx -104.65, \qquad U_T = 0 + 50c_T \approx 104.65 \quad \text{(kWh)}.$$

Interpretation. About 95% of electricity-use outcomes are within 104.65 kWh of forecast; 2.5% are further below and 2.5% further above.



3.4 Central sales deviations

Problem. A planning team describes how far sales falls below or rises above its target, in units. Use the chance pattern identified by the row marked 29 on page 2 of the supplied reference tables. Its middle represents exactly meeting the target. Each step of 1 in that pattern represents 20 units here. Amounts below the middle mean less than the target, and amounts above it mean more.

Questions/Tasks

C8 Find 2 amounts equally far below and above target enclosing the middle 80 percent of sales outcomes. Express the requested outcome using the reference pattern, work out the share at or below each boundary, find the boundaries in that pattern, and convert to the original units. Explain how the shares above or below the boundaries are used and interpret the result.

Worked solution

Translating the information into mathematics Let X_T be sales minus its target, in units, and let T be the value in the reference pattern. The specified row gives Student's t distribution with 29 degrees of freedom. The middle and conversion give

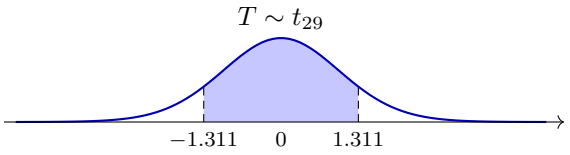
$$X_T = 0 + 20T, \qquad T = \frac{X_T - 0}{20}, \qquad T \sim t_{29}.$$

The location is 0 and the scale is 20 units. This scale is not automatically a population standard deviation.

$$P(-c < T < c) = 0.8000 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.1000.$$

Divide the outside complement $1 - p$ by 2 because symmetry makes the tails equal.

$$F_{29}(c) = 0.9000, \qquad c_T = F_{29}^{-1}(0.9) \approx 1.311.$$



The standardized endpoints are $[-1.311, 1.311]$.

$$L_T = 0 - 20c_T \approx -26.23, \qquad U_T = 0 + 20c_T \approx 26.23 \quad \text{(units)}.$$

Interpretation. About 80% of sales outcomes are within 26.23 units of target; 10% are further below and 10% further above.

3.5 2-tail checks on filling weights

Problem. A quality team repeatedly chooses 6 filling weights at random. Each choice is independent: knowing 1 result does not change the chances for another. The full group follows the bell-shaped chance pattern used on page 1 of the supplied reference tables, allowing for where the group is centered and how widely its values are spread. The full-group average exists, but its spread is unknown. For each chosen group, find its average and subtract the full-group average. To measure the spread within the chosen group, find how far each value is from its group average, multiply each difference by itself, add the results, divide by 1 fewer than the number chosen, and find the positive number which, multiplied by itself, gives that result. Divide this spread by the positive number which, multiplied by itself, gives 6, then divide the earlier difference between averages by that result to obtain a score. Find 2 boundaries equally far below and above 0 so that 10 percent of scores altogether fall below the lower boundary or above the upper boundary.

Questions/Tasks

C8 Express the requested outcome using the score, work out the share at or below each boundary, and find both boundaries. Explain how the share outside the boundaries is divided and interpret the result.

Worked solution

Translating the information into mathematics Let $X_1, \ldots, X_6$ be the independent random filling weights, from a normal population with mean μ and unknown standard deviation σ. Let $\bar{X}$ be the sample mean and S the sample standard deviation. The described spread and score are

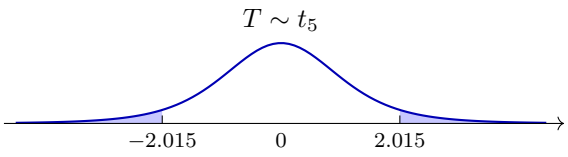
$$S^2 = \frac{\sum_{j=1}^{6}(X_j - \bar{X})^2}{5}, \qquad T = \frac{\bar{X} - \mu}{S/\sqrt{6}} \sim t_5.$$

Let $c > 0$ be the upper boundary; the lower boundary is $-c$.

$$P(|T| > c) = 0.1000 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0500.$$

Divide the combined outside area by 2 because symmetry makes the tails equal.

$$F_5(c) = 0.9500, \qquad c_T = F_5^{-1}(0.95) \approx 2.015.$$



The standardized endpoints are $[-2.015, 2.015]$. **Interpretation.** The t interval contains 90% of standardized sample-mean outcomes; 5% lie in each excluded tail.

3.6 2-tail checks on processing times

Problem. A quality team repeatedly chooses 25 processing times at random. Each choice is independent: knowing 1 result does not change the chances for another. The full group follows the bell-shaped chance pattern used on page 1 of the supplied reference tables, allowing for where the group is centered and how widely its values are spread. The full-group average exists, but its spread is unknown. For each chosen group, find its average and subtract the full-group average. To measure the spread within the chosen group, find how far each value is from its group average, multiply each difference by itself, add the results, divide by 1 fewer than the number chosen, and find the positive number which, multiplied by itself, gives that result. Divide this spread by the positive number which, multiplied by itself, gives 25, then divide the earlier difference between averages by that result to obtain a score. Find 2 boundaries equally far below and above 0 so that 4 percent of scores altogether fall below the lower boundary or above the upper boundary.

Questions/Tasks

C8 Express the requested outcome using the score, work out the share at or below each boundary, and find both boundaries. Explain how the share outside the boundaries is divided and interpret the result.

Worked solution

Translating the information into mathematics Let $X_1, \ldots, X_{25}$ be the independent random processing times, from a normal population with mean μ and unknown standard deviation σ. Let $\bar{X}$ be the sample mean and S the sample standard deviation. The described spread and score are

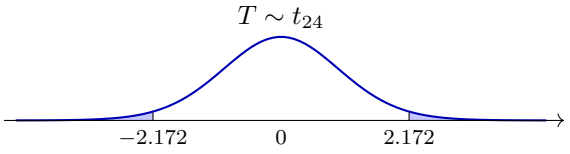
$$S^2 = \frac{\sum_{j=1}^{25}(X_j - \bar{X})^2}{24}, \qquad T = \frac{\bar{X} - \mu}{S/\sqrt{25}} \sim t_{24}.$$

Let $c > 0$ be the upper boundary; the lower boundary is $-c$.

$$P(|T| > c) = 0.0400 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0200.$$

Divide the combined outside area by 2 because symmetry makes the tails equal.

$$F_{24}(c) = 0.9800, \qquad c_T = F_{24}^{-1}(0.98) \approx 2.172.$$



The standardized endpoints are $[-2.172, 2.172]$. **Interpretation.** The t interval contains 96% of standardized sample-mean outcomes; 2% lie in each excluded tail.